# Physics-Grounded Rapid Grasp Pose Synthesis for Novel Objects

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Robotic grasping models trained on large-scale datasets have achieved remarkable performance on common objects but often fail when encountering out-of-distribution (OOD) objects or non-standard grippers due to dataset bias and poor physical grounding. To address this, an efficient framework is proposed for fine-tuning grasping models on novel objects. High-quality 3D object meshes are reconstructed using Neural Radiance Fields (NeRF), grasp labels are generated rapidly via force-closure analysis, and a physics-based dynamic filtering process is introduced to discard unstable grasps. In a randomized simulation benchmark of 25 objects, including six held-out objects, fine-tuning with the proposed labels raises the top-1 grasp success rate from 62% to 91% (2,500 trials per method). In real-robot experiments on nine objects (135 trials per method), the success rate increases from 29.6% to 73.3%. Generating 6,500 grasp labels takes about 2.2 h, compared with about 13 h for an antipodal sampling pipeline. The framework provides a scalable grasp label generation pipeline and a physics-grounded filtering mechanism that enhances real-world grasping robustness.

## 1 Introduction

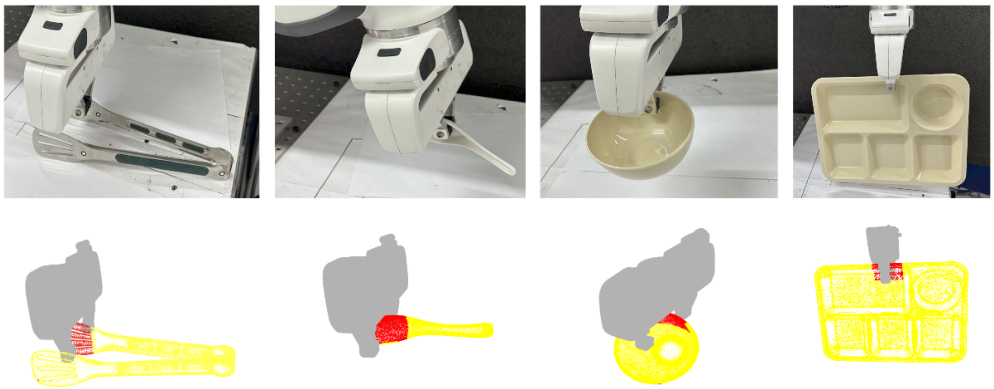


Figure 1: **The grasping region of the hand.** The generated closing region varies across objects and is necessary for point-cloud-based grasping algorithms.

Grasping is one of the most fundamental skills in prehensile manipulation, enabling robots to actively interact with their environment and accomplish a wide range of complex tasks. Recent years have witnessed an intense shift toward foundation grasp models: large neural networks trained on millions of synthetically rendered or real RGB-D point-cloud scenes paired with dense 6-Degree-of-Freedom (DoF) ground-truth grasp labels, such as FoundationGrasp [1], GraspLDM [2], and AnyGrasp [3]. These models learn object-agnostic geometric and grasping features during pre-training and, at inference, output the grasp poses that are most likely to succeed within milliseconds. They reach human-level performance on standard benchmarks of household items and exhibit strong zero-shot transfer to novel conditions. This data-driven approach marks a step change from analytic grasp planners [4] to foundation models for robust, real-world manipulation pipelines.

Despite their impressive breadth, current deep-learning grasp models rely on the diversity and coverage of offline datasets, which makes them highly vulnerable to out-of-distribution (OOD) novelties, such as special grippers with non-standard fingertip shapes or novel objects with unusual materials. In such cases, the grasp success rate can drop dramatically. For example, RoboReflect [5] evaluates AnyGrasp on eight "ambiguous-condition" everyday objects (e.g., partly empty tissue bags, open-lid cups, and ice-cream bars where only the stick is graspable) and reports low success rates on several of these OOD conditions. Furthermore, prior work shows that learning-based grasp models not only fail significantly

on OOD objects but also exhibit structural bias toward certain failure modes. Tang et al. [1] argue that limited generalization to novel elements outside the distribution is a natural consequence of the closed-set nature of the prior knowledge used.

To resolve this problem, a common remedial practice is to fine-tune the model on a small-scale, high-quality, domain-specific labeled dataset that fits the distribution of the novelties. While effective, this practice involves intensive data collection and labeling effort.

Current pipelines, such as antipodal pose sampling and analytic metrics [6], generate grasps from the contact points of parallel jaws on the object by performing collision detection and random sampling of gripper poses within a certain angular range. The core of these generation pipelines is to produce the closing region of the hand (see **Figure 1**) and the corresponding score annotations. However, using the antipodal method [7] to generate grasps for novel objects is computationally expensive (e.g., about 13 h for 6,500 grasp annotations on a 13th-generation Intel Core i7 CPU). Additionally, the two fingers of the gripper may not contact the object simultaneously, which shifts the object during closing. Existing pipelines may therefore assign scores that ignore execution dynamics. As a result, the fine-tuned model inherits noisy supervision and offers only modest OOD gains.

To address these challenges, the goal is reformulated as an offline pipeline that produces reliable grasp annotations for OOD objects and uses them to adapt a base grasping model. A physics-informed grasp-label pipeline is introduced that cuts label generation time by a factor of six and filters out dynamically unstable grasps before they reach the network. Specifically, rapid 3D mesh reconstruction of real-world objects is first performed with NeRF. Symmetry-aware force-closure grasp poses are then generated. Finally, a GPU-accelerated physics grounding module filters out grasp candidates that slip, violate torque limits, or induce large center-of-mass shifts.

Two distinct uses of simulation are reported in this paper and should not be confused. (i) *Label filtering*: the physics module executes every candidate grasp in simulation before training; applied to 6,500 antipodal candidates for each of 13 YCB objects [8], this amounts to 84,500 simulated executions, of which 42.2% are rejected. (ii) *Model evaluation*: the fine-tuned model is then tested in a separate simulated benchmark with randomized friction, mass, and object pose on 25 objects, six of which are never used for fine-tuning. In this evaluation (2,500 trials per method), fine-tuning with the proposed labels raises the top-1 grasp success rate from 62% to 91%; the physics filtering alone contributes 10.6 percentage points (80.4% to 91.0%), and on the six held-out objects the success rate rises from 66.7% to 88.3%. In real-robot experiments on nine objects (135 trials per method), the success rate improves from 29.6% to 73.3%.

The main contributions are as follows:

- Coverage-guided candidate sampling around the gripper closure region.
- Physics-grounded dynamic re-scoring, including force closure and collision avoidance during approach and closing.
- A practical one-shot offline data pipeline, from a small set of views via neural reconstruction, that improves real-robot success without adding runtime overhead.

## 2 Related Work

### 2.1 Analytical Grasp Pose Generation

Early work on grasp-pose generation was dominated by statistical and analytic planners that treat grasping as a search in wrench space under idealized rigid-body assumptions. Classic force-closure and Grasp Wrench Space (GWS) metrics were embedded in probabilistic frameworks that learn generative shape models or plan under configuration uncertainty, such as probabilistic geometry priors for class-level grasp planning and Bayesian planners that propagate pose and friction uncertainty during planar pushing grasps. These methods provide stability guarantees but require watertight meshes, precise inertial parameters, and exhaustive sampling, making them brittle in cluttered or sensor-noisy scenes. Goldfeder

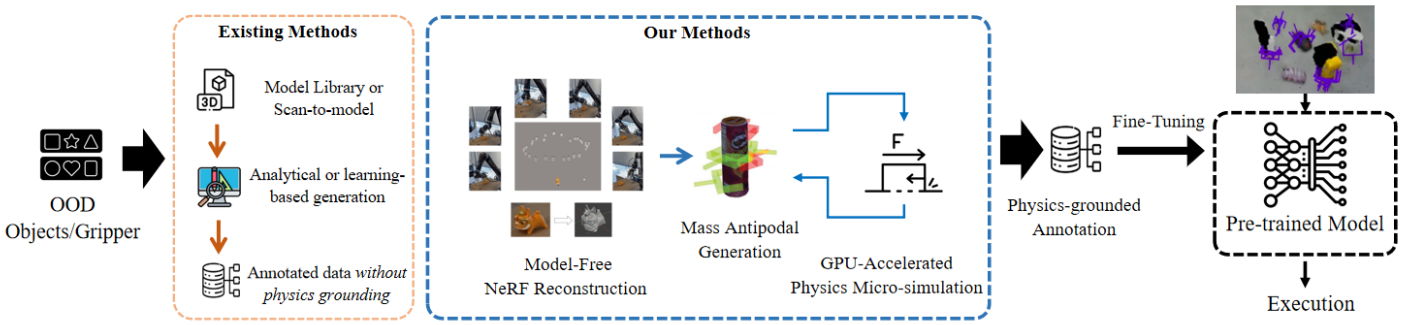



Figure 2: **Framework.** The workflow first generates grasp configurations rapidly through geometric analysis, then eliminates infeasible grasps via large-scale simulation, and finally outputs a rescored grasp dataset for fine-tuning.

et al. [9] created the Columbia Grasp Database using GraspIt! [10] combined with form-closure analysis. Generating each grasp takes about 12 s (**Table 2**), and building the full dataset takes about one month on a server. This computational cost is too high for fine-tuning.

To reduce sampling cost, researchers introduced heuristic pipelines that exploit geometric cues. Antipodal sampling of surface-point pairs remains the most popular method, while faster variants such as QuickGrasp [11] project points onto PCA axes to yield near-antipodal pairs in milliseconds. Although computationally light, such heuristics ignore object dynamics. Analytically good grasps frequently slip when the fingers touch the object sequentially or when torque limits are violated.

## 2.2 Learning-Based Grasp Pose Generation

The past decade has seen a shift toward learning-based generation that treats grasping as a data-driven prediction problem. Early CNNs predicted planar rectangles from RGB-D images [12]; PointNetGPD [13] generalized this idea to 6-DoF by scoring antipodal proposals with a PointNet backbone [14]. These methods were used to construct large-scale datasets such as GraspNet [15], which underlie large foundation grasping models such as AnyGrasp [3]. Methods like PointNetGPD consist of an offline and an online stage. In the offline stage, grasps are sampled on each object point cloud with the antipodal method [7], and each grasp corresponds to a closing region of the hand and a score annotation. A PointNet-based network [14] is then trained to predict this score. Several problems persist. First, when grasping novel objects or using gripper designs that differ from those in the training dataset, the success rate is much lower. Second, offline generation of the closing regions for one object requires more than ten hours on a mainstream desktop computer (13th-generation Intel Core i7), which is too time-consuming. Third, the score annotation is based on force closure and does not account for object motion during gripper closing, which can cause failure. Thus, while these learning-based models excel on in-distribution objects, they still require high-quality, domain-specific data to adapt to unusual grippers or materials.

## 2.3 Grasping Pose Candidate Scoring for Few-Shot Learning Labels

The core of grasp pose annotation is to evaluate grasp quality, namely the probability of grasp success. In OOD tasks, where large synthetic datasets lack domain specificity and learned generators inherit dataset biases, analytical methods are often used to label domain-specific data for fine-tuning. Analytical methods evaluate grasp quality from the geometry of the gripper configuration and the object. Force closure [16] and GWS [17] are two mainstream metrics. The force-closure method considers friction between the gripper and the object, while GWS can also handle frictionless situations. These methods are versatile and have driven significant progress in grasp research. However, they do not account for the fact that, during execution, other points on the gripper may contact the object first, causing object

motion and the failure of grasps that are rated highly. This motivates the scoring strategy introduced here.

### 2.4 Grasp Dataset Generation and Stability Evaluation

Large-scale grasp datasets are typically produced by sampling candidates on object meshes and labeling them with analytic metrics or simulated outcomes. Dex-Net 2.0 [6] labels parallel-jaw grasps with robust analytic quality metrics, whereas Eppner et al. [18] execute more than one billion grasps in physics simulation and show that the choice of sampling scheme strongly affects grasp coverage. EGAD [19] emphasizes geometric diversity of the object set to make grasp evaluation reproducible. Niu et al. [20] proposed a customizable 6-DoF grasping dataset together with an interactive training method for graph convolutional networks, enabling grasp labels to be tailored to specific objects and grippers. Beyond geometric and analytic labels, Niu et al. [21] collected a visual–tactile grasp dataset and introduced a grasp margin matrix to evaluate grasp stability, offering a sensing-based perspective on how close a grasp is to failure. The present work is complementary to these efforts: instead of relying on tactile sensing or billions of simulated trials, it targets fast, object-specific label generation for fine-tuning and uses simulated closing and lifting dynamics as a proxy for grasp stability. Tactile stability measures such as the grasp margin matrix [21] could provide an independent real-world validation of the proposed physics-based score.

## 3 Methods

Grasp poses are represented by a 6-DoF pose consisting of a 3D translation and an orientation. Surface normals are computed from the reconstructed mesh. Initial candidates are sampled to cover the gripper closure region on the object surface. For each candidate, a physics-grounded score is computed that combines: (a) a force-closure term, where a lower required friction coefficient implies a higher score; (b) collision checks along the approach and closing trajectories; and (c) stability terms under expected disturbances. Candidates that collide or fail the stability test are removed, and the remaining candidates are re-weighted. All of these computations are executed offline. The parameter values used in this work are summarized in **Table 1**.

### 3.1 Generate CAD Models of Novel Objects

Because data-driven grasping algorithms may fail on some novel objects when applied to new scenes, they need to be fine-tuned for these objects. Neural reconstruction is used to build a geometric proxy of each object for candidate generation. The proposed pipeline is complementary to existing systems, as it supplies higher-quality labels for targeted objects.

The first requirement is a CAD model of each novel object. However, creating a detailed model manually is cumbersome. A framework is therefore designed that requires only a robotic arm and an RGB-D camera to produce the CAD model of a new object for subsequent grasp generation.

First, the robotic arm positions the camera around the object, using the depth returned by the camera to set the viewing distance (see **Figure 2**), and captures 32 views of the object from different angles. These images are used to train a NeRF [22] for approximately 30k iterations. A surface mesh is then extracted with marching cubes [23] on a $256^3$ grid and upsampled to obtain a denser point set for subsequent grasp generation.

### 3.2 Generate Grasp

Generating grasp datasets with antipodal methods, as in the Columbia Grasp Database [9] and PointNetGPD [13], is time-consuming: thousands of grasps can take more than ten hours. Therefore, an efficient grasp generation pipeline is proposed to enable rapid fine-tuning on novel objects.

First, the grasp configuration is defined as

$$G = [p, r] \in \mathbb{R}^6 \tag{1}$$

where $p = [x, y, z] \in \mathbb{R}^3$ is the grasp point (the midpoint between the two fingertips) and $r = [r_x, r_y, r_z] \in \mathbb{R}^3$ is the gripper orientation.

Antipodal methods generate grasps from contact points. Specifically, a point on the object is selected, a second contact point is searched within a sphere whose radius equals the gripper width, and grasp poses are then sampled within a certain angular range around the contact pair. The grasp poses are evaluated for collision and force-closure scores. Because each contact pair can be found starting from either of its two points, this process produces many repeated grasp configurations for parallel jaws, for example

$$G\left[\frac{n_1 + n_2}{2}, r\right] = G\left[\frac{n_2 + n_1}{2}, r\right] \tag{2}$$

where $n_1$ and $n_2$ are the two contact points.

Since the aim is to obtain diverse closing regions of the hand for fine-tuning while accelerating generation, a framework different from prior approaches [15] is adopted. A fixed set of grasp orientations is sampled uniformly in advance,

$$r_i \in \{r_1, r_2, \ldots, r_N\} \tag{3}$$

and grasps are generated by iterating over every point on the object with each orientation. Approach directions are taken from a Fibonacci lattice [24] of 256 directions, and for each approach direction 12 rotations about the approach axis are added, giving $N = 3{,}072$ orientations. Neighboring approach directions are about 12° apart, and the rotations about the approach axis are spaced at 30°.

The grasp generation procedure is summarized in Algorithm 1.

**Algorithm 1** Generate Grasp

**Require:** CAD model, configuration of the parallel jaws
**Ensure:** $G$ [grasp point, grasp configuration, grasp quality]

```
Define N uniformly sampled grasp orientations r_i
Reorient the point-cloud normals
for each of the n points on the object do
    if the contact normals lie inside the friction cone and the gripper does not collide with the object then
        add g_i to G
    end if
end for
Remove repeated closing regions
Compute the force-closure grasp quality
```

In this process, collision detection and local reorientation of the surface normals ensure that the generated grasps do not collide with the object. Candidates whose contact normals deviate from the closing direction by more than the friction-cone half-angle $\arctan \mu$ (about 27° for $\mu = 0.5$) are discarded. Because the orientation set is fixed, the sampling process is $O(n)$ in the number of object points, compared with $O(n^2)$ for antipodal sampling over point pairs. Duplicate closing regions are then removed: two grasps are treated as duplicates if their translations differ by less than 5 mm and their orientations differ by less than 15°. Finally, the same force-closure scoring strategy as PointNetGPD [13] and GraspNet [15] is applied (Equation 8). For each object, 10,000 candidate grasps are kept after deduplication and passed to the physics module. As shown in **Table 2**, the method is substantially faster than antipodal sampling, while producing the same force-closure score annotations as previous pipelines; the effect of the resulting labels on grasp success is evaluated in Section 4.4.

The method is faster than the antipodal approach yet produces labels of the same form for three reasons. First, it does not generate duplicate grasps. Second, unreasonable grasps are filtered out using the surface normals estimated from the object point cloud (see **Figure 3**), which is cheaper than eliminating them with a force-closure test. Third, the same scoring and labeling method as the antipodal approach is used, ensuring consistency of the generated grasp data and scores.

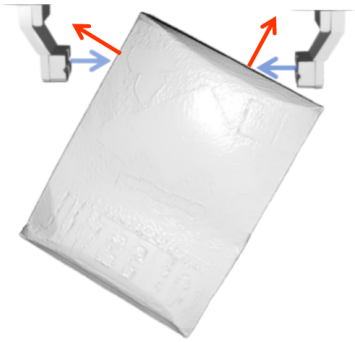


Figure 3: Grasps whose contact-force directions fall outside the friction cone are quickly filtered out using surface normals estimated from the object point cloud.

### 3.3 Stable Grasp

When executing a grasp, the gripper moves to the grasp point along the approach direction and then closes. An important effect of this closing step is often overlooked. As shown in **Figure 4**, if one finger touches the object before the other, the force balance of the object is disrupted, which moves the object and results in an unstable grasp.

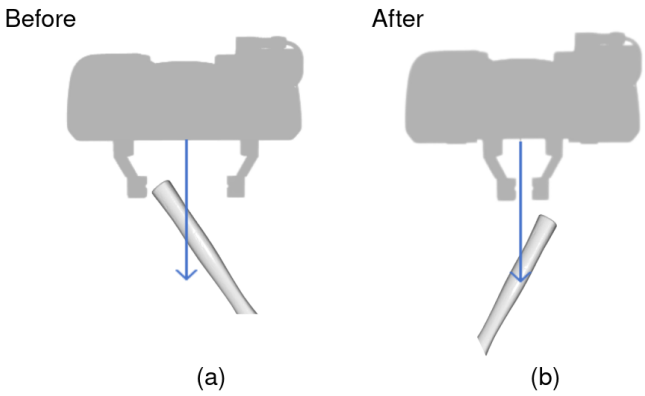



Figure 4: **Statically generated grasps.** (a) The state before the gripper closes, where the closing region is not symmetric with respect to the grasp approach. (b) The state after the gripper closes, where the grasp fails because the left finger touches the object first and displaces it.

**Closing symmetry.** All quantities below are expressed in the gripper frame, whose origin is the grasp point $p$ and whose unit vector $u$ is the finger closing direction. The reflection across the plane that bisects the two fingers is

$$R_m = I - 2uu^T \in \mathbb{R}^{3\times 3} \tag{4}$$

Let $PC_1$ and $PC_2$ denote the object points inside the closing region on the two sides of this plane. A grasp is perfectly symmetric if

$$R_m\, PC_1 = PC_2 \tag{5}$$

in which case both fingers make contact at the same instant and the net closing impulse on the object is zero. In practice, the deviation from Equation 5 is measured by the free travel of each finger before contact,

$$d_k = \min_{q\in PC_k} \left( \frac{w}{2} - |u^T q| \right), \quad k = 1, 2 \tag{6}$$

where $w$ is the maximum opening width of the gripper. A symmetric grasp has $d_1 = d_2$. The asymmetry

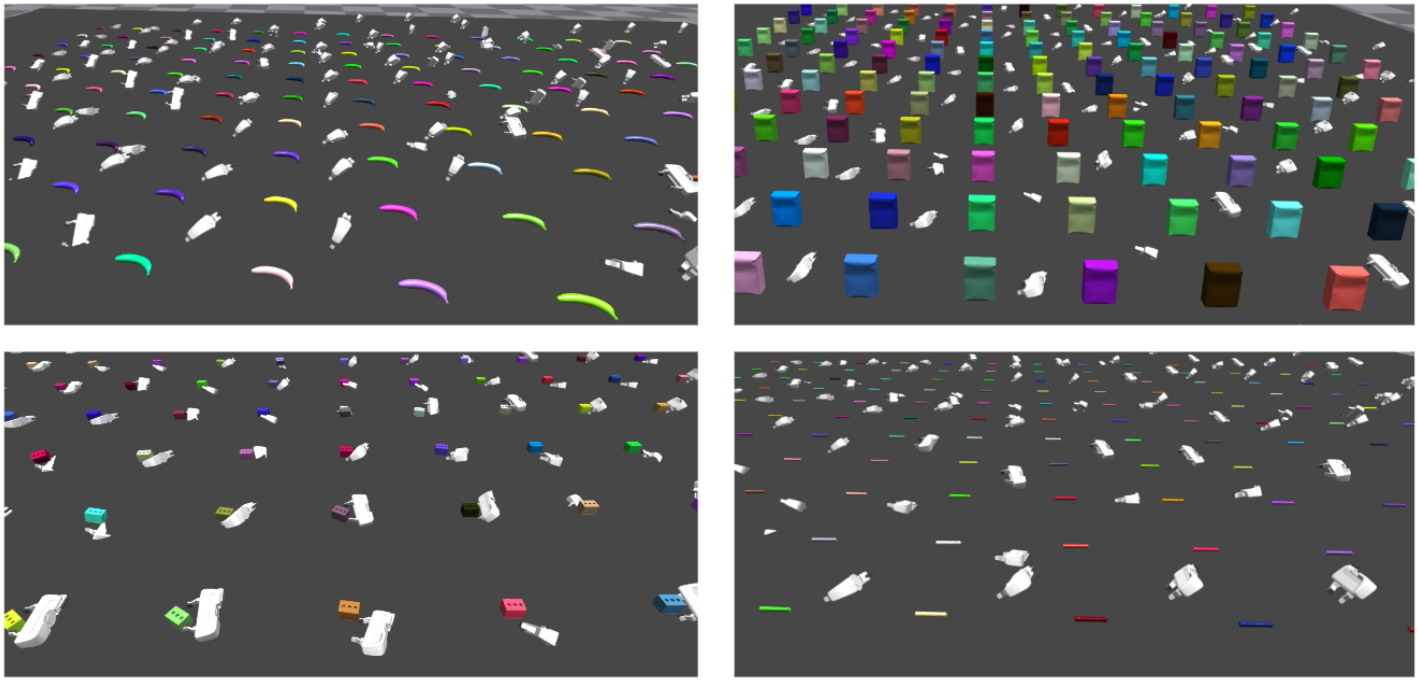


Figure 5: **Screening.** Grasp configurations are screened using large-scale simulation [25].

weight is

$$w_{\text{sym}} = 1 - \frac{|d_1 - d_2|}{w} \in [0, 1] \tag{7}$$

which equals 1 when both fingers touch simultaneously and decreases linearly with the difference in contact time.

**Static force-closure score.** Following PointNetGPD [13] and GraspNet [15], the static score of a grasp is derived from the smallest friction coefficient $\mu_{\text{min}} \in \{0.1, 0.2, \ldots, 1.0\}$ for which the two contacts achieve force closure:

$$s_{\text{fc}} = 1.1 - \mu_{\text{min}} \in [0.1, 1.0] \tag{8}$$

A grasp that needs less friction to hold the object therefore receives a higher score.

**Dynamic stability.** Grasps that are good under static force closure can still fail during execution because of asymmetric contact or insufficient gripping force. Each candidate is therefore executed in simulation, and the object pose is compared before and after the lift-and-hold test. Let $\Delta p$ be the displacement of the object's center of mass relative to the gripper and $\Delta\theta$ the change in object orientation. A grasp is considered successful if

$$\sigma = \mathbb{1}\left[\Delta p \leq \delta_p \ \wedge \ \Delta\theta \leq \delta_\theta \ \wedge \ \text{object not dropped}\right] \tag{9}$$

with $\delta_p = 1$ cm and $\delta_\theta = 15°$. For successful grasps, the stability weight is the center-of-mass displacement normalized by the gripper width,

$$w_{\text{stab}} = 1 - \frac{\Delta p}{w} \in [0, 1] \tag{10}$$

**Final grasp score.** The label used for fine-tuning combines the three terms:

$$s = \sigma \cdot s_{\text{fc}} \cdot w_{\text{sym}} \cdot w_{\text{stab}} \tag{11}$$

Grasps with $\sigma = 0$ are removed from the dataset. All factors lie in $[0, 1]$, so $s \in [0, 1]$ and the final score is never higher than the static force-closure score. A grasp keeps a high score only if it satisfies force closure, closes symmetrically, and holds the object without significant motion.

**Contact-force limit.** Grasps can also fail because the object slips out of the fingers. This failure arises from the limited contact force of the gripper: the two contacts can only resist a bounded moment. The grasp fails when the moment required to hold the object exceeds this bound,

$$\| r_c \times m\,(g - a) \| > \tau_{\max}(\mu, F_{\max}) \tag{12}$$

where $r_c$ is the vector from the grasp point to the object's center of mass, $m$ is the object mass, $g$ is the gravitational acceleration, $a$ is the acceleration imposed by the lifting and shaking motion, and $\tau_{\max}$ is the largest frictional moment the contacts can provide with friction coefficient $\mu$ and contact-force limit $F_{\max}$. Once the object starts to rotate, static friction turns into kinetic friction and the slip accelerates. Equation 12 is not evaluated in closed form because $\tau_{\max}$ depends on the contact patch geometry. It is instead checked implicitly by the simulation, in which each object is given its real mass and the gripping force is capped at $F_{\max}$.

During large-scale simulation, each grasp configuration is tested to determine whether the object can be lifted to the target position and held after the gripper closes. Grasp configurations that cannot hold the object within the contact-force limit receive $\sigma = 0$ and are filtered out. In this way, the retained grasp configurations are those more likely to succeed in practice. The simulation is performed in Isaac Gym [25] (**Figure 5**) with the model of the Franka Hand, the same parallel-jaw gripper that is used in the real-world experiments.

# 4 Experiments

## 4.1 Implementation Details

All parameters of the label generation pipeline are listed in **Table 1**. The object masses are taken from the official YCB data [8] for YCB objects and measured with a scale for the custom objects. In the filtering simulation, the gripper closes at 0.02 m/s until contact, lifts the object by 20 cm, and holds it for 3 s while a sinusoidal shaking disturbance of ±3 cm at 1 Hz is applied. The object pose is then compared with its pose at the end of closing to compute $\Delta p$ and $\Delta\theta$.

Table 1: Parameters of the label generation and filtering pipeline.

| Item | Setting |
|---|---|
| NeRF reconstruction | 32 views; approximately 30k iterations; marching cubes at $256^3$ |
| Orientation sampling | 256 Fibonacci-lattice approach directions × 12 rotations about the approach axis ($N$ = 3,072; approach directions about 12° apart, 30° roll steps) |
| Normal filter threshold | Friction-cone half-angle $\arctan\mu$ ($\approx 27°$ for $\mu = 0.5$) |
| Duplicate removal | Translation < 5 mm and rotation < 15° |
| Candidates per object | 10,000 |
| Friction coefficient $\mu$ | 0.5 |
| Object mass | Official YCB values; measured with a scale for custom objects |
| Contact-force limit $F_{\max}$ | 25 N |
| Gripper closing speed | 0.02 m/s |
| Simulator | Isaac Gym [25], $\Delta t = 1/60$ s, 4 substeps, Franka Hand |
| Lift-and-hold test | Lift 20 cm, hold 3 s, shaking ±3 cm at 1 Hz |
| Failure threshold | $\Delta p > 1$ cm, $\Delta\theta > 15°$, or object dropped |

## 4.2 Grasp Generation Efficiency

The proposed method accelerates grasp generation substantially compared with existing antipodal sampling pipelines. Using the average time per pose in **Table 2**, generating 6,500 grasp labels takes $6{,}500 \times 1.2$ s $\approx 2.2$ h with the proposed method, compared with $6{,}500 \times 7.2$ s $= 13$ h with the antipodal pipeline of PointNetGPD, a speed-up of 6× (**Table 3**). With the default setting of 10,000 labels per object, the proposed pipeline takes about 3.3 h.

Table 2: Grasp generation cost of different label generation pipelines.

| Methods | Average quantity | Average speed (per pose) |
|---|---|---|
| PointNetGPD [13] | 6500 | 7.2 s |
| Columbia grasp [9] | 5000 | 12 s |
| Fast Grasp [26] | 2000 | 0.5 s |
| BWG [18] | 15000 | 240 s |
| Ours | 10000 | 1.2 s |

Grasp orientations are sampled in advance to generate the grasp candidates, which are then annotated with the same force-closure method as traditional pipelines. As shown by the original force-closure scores in **Figure 6** (purple), the grasp score distributions remain similar across different novel objects, even though their shapes and characteristics vary. This indicates that the method is consistent and can be applied to various types of objects.

Table 3: Time to generate 6,500 grasp labels on a 13th-generation Intel Core i7 CPU, computed from the average time per pose in Table 2.

| Method | Time per pose | Total time | Speed-up |
|---|---|---|---|
| Antipodal sampling (PointNetGPD [13]) | 7.2 s | 13.0 h | 1× |
| Ours | 1.2 s | 2.2 h | 6× |

## 4.3 Physics-Based Label Filtering

This subsection reports the simulation used to *filter labels*. It is separate from the evaluation of the fine-tuned model in Section 4.4.

### 4.3.1 Asymmetric Closing

Large-scale grasp simulation quickly filters out grasp configurations that fail when the gripper closes. These failures are mostly caused by asymmetry of the closing region relative to the grasp approach: one finger contacts the object first, disrupts its equilibrium, and moves it. This effect is expected to be even stronger in cluttered scenes, where object motion during closing can also disturb neighboring objects; cluttered scenes are not evaluated in this work.

To filter the candidates, $n$ parallel simulation environments are created in which the object is placed at the same position and orientation. One grasp is executed in each environment, and the motion of the object during closing, lifting, and holding is recorded. If the displacement or orientation change of the object exceeds the threshold in Table 1, the grasp configuration is considered invalid and removed. The scores of the remaining grasps are re-weighted according to Equation 11.

As shown in **Figure 6**, the original force-closure score does not account for the motion of the object during grasping, and many grasps with high static scores receive low scores after re-evaluation.

### 4.3.2 Contact-Force-Limited Grasp

This filtering step primarily eliminates grasp configurations that cannot balance the moment of Equation 12 with the limited contact force of the gripper. If the contact force cannot provide sufficient frictional moment to keep the object stable, the object slides; static friction then turns into kinetic friction, which reduces the frictional force and accelerates the slip until the object may fall from the gripper.

With the contact-force limit set to 25 N, 6,500 antipodal grasp candidates were executed for each of 13 YCB objects, i.e., 84,500 simulated executions in total. Of these, 42.2% (35,659/84,500) failed after gripper closing and were removed from the label set. These 84,500 executions are used only to clean the training labels; they are not a measure of the success rate of the fine-tuned model.

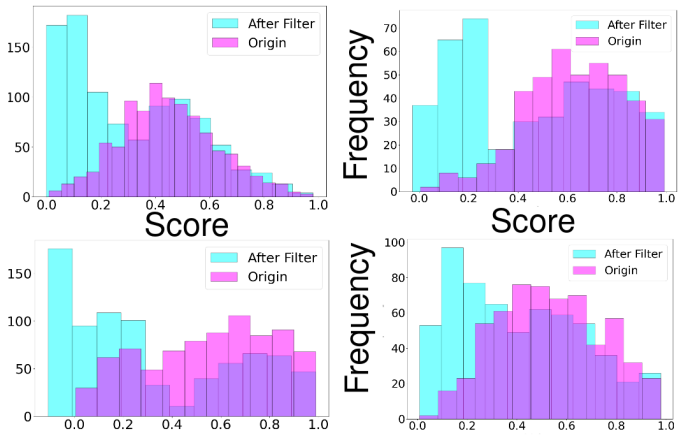



Figure 6: **Rescore.** Scores for four objects. Purple: original annotations based on force closure; cyan: re-evaluated scores after physics-based filtering. Many grasps with high original scores would not perform well in practice because of collisions during closing.

## 4.4 Simulated Evaluation and Ablation

**Protocol.** To measure the success rate of the fine-tuned model with sufficient statistical power, and to separate the contribution of each component, a large-scale evaluation is performed in Isaac Gym [25]. This evaluation is independent of the label filtering: the labels were filtered with a fixed friction coefficient, nominal masses, and a single object pose, whereas the evaluation randomizes all three and includes objects that were never used for fine-tuning. This reduces the risk that the reported gains merely reflect overfitting to the filtering conditions. The evaluation uses 25 objects in three groups: (i) the 13 YCB objects used for fine-tuning; (ii) the six OOD objects used for fine-tuning and in the real-world experiments (gamepad, glue bottle, three dolls, and tape); and (iii) six held-out YCB objects that were never used for fine-tuning, to assess generalization and rule out overfitting: mug, power drill, scissors, hammer, extra-large clamp, and Rubik's cube. For each object, 100 random stable resting poses are generated. In every trial, the friction coefficient is drawn uniformly from $[0.4, 1.0]$ and the object mass is scaled by a random factor in $[0.8, 1.2]$. The top-1 grasp predicted by the model is executed, and the trial counts as a success if the object is lifted by 20 cm and held for 3 s, the same criterion as in the real-world experiments. All other simulation settings, namely the Franka Hand model, closing speed (0.02 m/s), contact-force limit (25 N), and time step (1/60 s with 4 substeps), are identical to those of the filtering stage in Table 1; only the friction, mass, and object pose are randomized. Each method is thus evaluated on $25 \times 100 = 2{,}500$ trials, for 10,000 trials in total.

**Compared methods.** Four variants of the same base model (PointNetGPD [13]) are compared: (1) no fine-tuning; (2) fine-tuning with antipodal labels; (3) fine-tuning with the proposed labels without physics-based filtering, i.e., candidates scored by $s_{\text{fc}}$ only; and (4) the full method, i.e., fine-tuning with labels scored by Equation 11. Variants (3) and (4) differ only in the physics filtering module and therefore isolate its contribution.

Table 4: Top-1 grasp success rates in the randomized Isaac Gym evaluation (100 trials per object). Values in brackets are 95% Wilson confidence intervals [27].

| Method | Seen YCB (13 obj., 1,300 trials) | Seen OOD (6 obj., 600 trials) | Held-out (6 obj., 600 trials) | Overall (25 obj., 2,500 trials) |
|---|---|---|---|---|
| No fine-tuning | 70.0% [67.5, 72.4] | 40.0% [36.2, 44.0] | 66.7% [62.8, 70.3] | 62.0% [60.1, 63.9] |
| Antipodal labels | 83.1% [80.9, 85.0] | 63.3% [59.4, 67.1] | 71.7% [67.9, 75.1] | 75.6% [73.9, 77.2] |
| Ours w/o physics filtering | 86.9% [85.0, 88.6] | 71.7% [67.9, 75.1] | 75.0% [71.4, 78.3] | 80.4% [78.8, 81.9] |
| Ours (full) | **95.0%** [93.7, 96.1] | **85.0%** [81.9, 87.6] | **88.3%** [85.5, 90.7] | **91.0%** [89.8, 92.1] |

**Results.** As shown in **Table 4**, the full method raises the overall top-1 success rate from 62.0% without fine-tuning to 91.0%. With 2,500 trials per method, the half-width of the 95% confidence interval is

about 1.9 percentage points at 62% and 1.1 percentage points at 91%, so the 29-point gain is far larger than the sampling uncertainty. Fine-tuning with antipodal labels already improves the overall success rate to 75.6%, and replacing them with the proposed force-closure labels without physics filtering raises it further to 80.4%.

**Ablation.** Variants (3) and (4) differ only in the physics filtering module. Adding the filter raises the overall success rate from 80.4% to 91.0% (+10.6 percentage points; two-sided Fisher's exact test, $p < 10^{-20}$), so it accounts for about one third of the total 29-point gain over the base model. The gain is consistent across all three groups: +8.1 points on the seen YCB objects (86.9% to 95.0%), +13.3 points on the seen OOD objects (71.7% to 85.0%), and +13.3 points on the held-out objects (75.0% to 88.3%), each with $p < 10^{-7}$. The largest gains occur on the OOD and held-out objects. Many of these objects have irregular shapes or off-center mass distributions (e.g., the dolls, the power drill, the hammer, and the mug), for which grasps that pass the static force-closure test but slip or rotate during closing and lifting are more common; such grasps are precisely the candidates removed by the physics filter.

**Generalization to held-out objects.** On the six held-out objects, which were never used for fine-tuning, the full method reaches 88.3% compared with 66.7% without fine-tuning and 71.7% with antipodal labels. In contrast, the gains of the antipodal labels (+5.0 points, $p = 0.07$) and of the unfiltered proposed labels (+8.3 points) on these objects are much smaller than on the seen objects. Fine-tuning with physics-filtered labels therefore does not merely fit the fine-tuning set: the model appears to learn a more general preference for dynamically stable grasps that transfers to unseen geometries. Moreover, the success rate on the held-out objects (88.3%) is comparable to that on the seen OOD objects (85.0%), even though the evaluation randomizes friction, mass, and pose beyond the conditions used for filtering, which further indicates that the improvement does not stem from overfitting to the filtering conditions.

### 4.5 Real-World Experiments

**Setup.** The real-world experiments (**Figure 7**) use a Franka Research 3 robot arm with its standard Franka Hand parallel-jaw gripper and an Intel RealSense D435i RGB-D camera. Nine objects are tested: six OOD objects (gamepad, glue bottle, three dolls, and tape) and three YCB-like objects (box, milk box, and Coke can). All nine objects were used during fine-tuning; generalization to unseen objects is evaluated in simulation (Section 4.4).

**Protocol.** Each object is tested in 15 trials per method. In each trial, the object is placed alone on the table at a random position and orientation within the camera's field of view. The model receives the observed point cloud and the top-1 grasp is executed without retries. A trial counts as a success if the object is lifted by 20 cm and held for 3 s. Three methods are compared: PointNetGPD without fine-tuning, PointNetGPD fine-tuned with antipodal labels, and PointNetGPD fine-tuned with the proposed labels.

**Results.** As shown in **Table 5**, fine-tuning with the proposed labels raises the overall real-world success rate from 29.6% (40/135) without fine-tuning and 52.6% (71/135) with antipodal labels to 73.3% (99/135). Both improvements are statistically significant (two-sided Fisher's exact test, $p < 10^{-12}$ and $p < 0.001$, respectively). Before fine-tuning, all attempts to grasp the gamepad failed; after fine-tuning with the proposed labels, it was grasped in 14 of 15 trials. Fine-tuning did not degrade performance on the YCB objects. Because only 15 trials are performed per object, the per-object confidence intervals are wide (e.g., [10.9, 52.0]% for 4/15), and per-object differences should be interpreted with caution; the aggregated results and the large-scale simulated evaluation in Section 4.4 provide the statistically reliable comparison.

In an additional round, the model was fine-tuned on only five objects (the three dolls, the gamepad, and the milk box); each of the three dolls was then grasped successfully in all 15 trials. The three dolls have similar shapes, and their good grasp regions are also very similar. When the dolls make up most of the fine-tuning set, their labels reinforce one another, and the model learns these shared grasp regions more effectively than when the dolls are only a small part of a larger fine-tuning set.

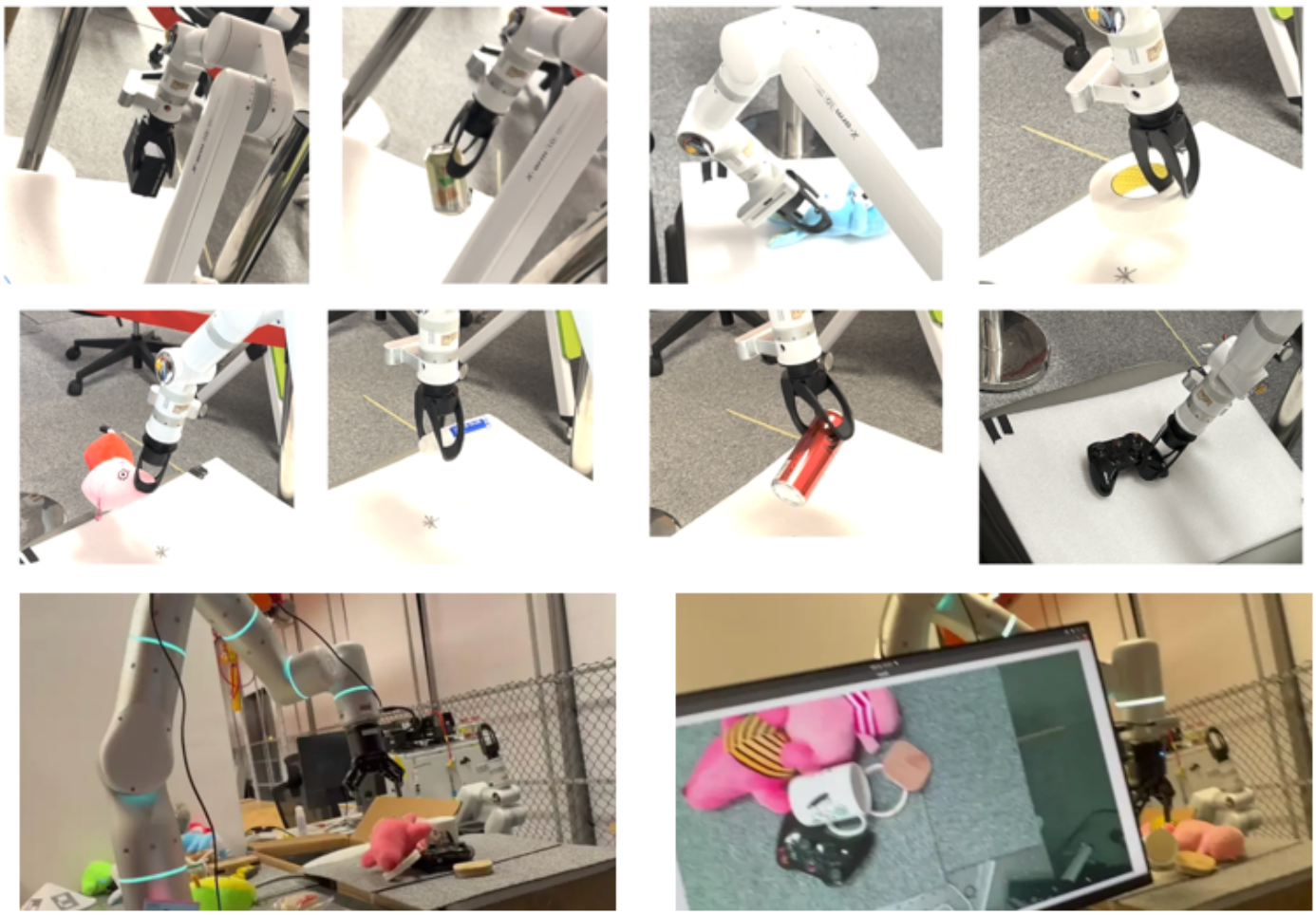


Figure 7: **Real-world experiment.** After fine-tuning, the model was tested in real-robot grasping experiments using the objects employed during fine-tuning. The experiments use a Franka Research 3 arm with the Franka Hand parallel-jaw gripper: the fingers open fully, the gripper moves to the grasp position, and the fingers then close simultaneously to perform the grasp.

Table 5: Real-world grasping success rates (successful grasps / attempts). The bottom rows give the totals over all nine objects and the 95% Wilson confidence intervals [27].

| Group | Object | PointNetGPD (w/o FT) | PointNetGPD (FT, antipodal) | Ours |
|---|---|---|---|---|
| OOD objects | Gamepad | 0/15 | 0/15 | 14/15 |
| | Glue bottle | 5/15 | 12/15 | 13/15 |
| | Doll_01 | 0/15 | 5/15 | 6/15 |
| | Doll_02 | 0/15 | 0/15 | 5/15 |
| | Doll_03 | 4/15 | 10/15 | 12/15 |
| | Tape | 2/15 | 4/15 | 4/15 |
| | Subtotal | 11/90 (12.2%) | 31/90 (34.4%) | 54/90 (60.0%) |
| YCB objects | Box | 8/15 | 10/15 | 15/15 |
| | Milk box | 10/15 | 15/15 | 15/15 |
| | Coke can | 11/15 | 15/15 | 15/15 |
| | Subtotal | 29/45 (64.4%) | 40/45 (88.9%) | 45/45 (100%) |
| All | Total | 40/135 (29.6%) | 71/135 (52.6%) | 99/135 (73.3%) |
| | 95% CI | [22.6, 37.8] | [44.2, 60.8] | [65.3, 80.1] |

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## Conflicts of Interest

The authors declare no conflicts of interest.

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